Toxpipe Evaluation Process and Results

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1 Description

ToxPipe is evaluated in multiple levels with different sets of queries. The response at each stage was evaluated based on semantic similarity with the expected keywords or phrases.

2 Evaluation Sets

2.1 Levels

ToxPipe system is evaluated at three levels.

Base model

This level includes the following foundational models,

- o3
- GPT-5 (high reasoning)
- GPT-5 (low reasoning)
- GPT-5 Nano
- GPT-4o
- Claude 4.5 Haiku
- Claude 4.5 Sonnet
- Claude 3.7 Sonnet
- Gemini 2.5 Pro
- Gemini 2.5 Flash
- Llama 4 Scout 17B (Instruct)
- Mistral Large 2

RAG

In this level, the aforementioned foundational models are used with an extra RAG (Retrieval Augmented Generation) step.

MCP

In this level, the LLMs are exposed to the tools that leverage the internal databases of ToxPipe using MCP (Model-Chain-Prompt) approach.

Name	Description
literature_search	Given a query, return relevant academic and scientific papers from PubMed. Use this tool if the user requests a literature search. Args: query: The query to perform a literature search on. This should be a string of at least one character and at most 9999 characters. The query should be specific enough to yield relevant results. Returns: A string containing the results of the literature search.
rag_search	Given a query, return relevant toxicological information from publications from the National Toxicology Program (NTP) at https://ntp.niehs.nih.gov/publications. These reports are retrieved via retrieval-augmented generation (RAG). The publications include chemical, toxicity, and technical reports. This tool should be used if the user requests a literature search or a RAG search. Args:

Name	Description
	query: The query to perform a RAG search on. This should be a string of at least one character and at most 9999 characters. The query should be specific enough to yield relevant results. Returns: A string containing the results of the RAG search, which may include information from the model's training data and/or from NTP publications.
is_valid_smiles	Given a SMILES string, return whether or not it is a valid SMILES representation. Args: smiles: A SMILES string representing a chemical's structure. This should be a string of at least one character and at most 255 characters. Returns: A boolean indicating whether or not the given SMILES string is valid.
smiles_to_mol_weight	Given a SMILES string, return the average molecular weight in g/mol of the chemical. Args: smiles: A SMILES string representing a chemical's structure. This should be a string of at least one character and at most 255 characters. Returns: The average molecular weight in g/mol of the chemical.
smiles_to_name	Given a chemical's SMILES representation, return its preferred name. If an exact mapping could not be found, the most structurally similar chemical's name is returned instead. Args: smiles: A SMILES string representing a chemical's structure. This should be a string of at least one character and at most 255 characters. Returns: A list of strings, where each string is structured as follows: chemical_name
casrn_to_name	Given a chemical's CASRN, return its preferred name. Args: casrn: A string representing the CASRN number for a chemical. This should be a string of at least one character and at most 255 characters. Returns: A string representing the preferred name of the chemical corresponding to the given CASRN. If no chemical with the given CASRN is found, a string indicating that no chemical name could be obtained is returned instead.
name_to_canonical_smiles	Given the name of a chemical, return its canonical SMILES representation.

Name	Description
	Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns: A string representing the canonical SMILES representation of the chemical corresponding to the given name. If no chemical with the given name is found, a string indicating that no SMILES could be obtained is returned instead.
ctd_chemical_to_genes	Given the name of a chemical and a species, return that chemical's associated gene interactions from the Comparative Toxicogenomics database (CTD). This tool returns a list of strings, where each string is an interaction for the given chemical in the specified species passed to the tool. Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. species: A string representing the species for which gene interactions are requested. Must be exactly one of: Homo sapiens, Mus musculus, Rattus norvegicus. Returns: A list of strings, where each string is an interaction for the given chemical in the specified species. If no interactions are found for the given chemical in the specified species, an empty list is returned.
ctd_chemical_to_diseases_direct	Given the name of a chemical, return that chemical's associated diseases with direct evidence (i.e., from a marker) from the Comparative Toxicogenomics database (CTD). Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns: A list of strings, where each string is a disease associated with the given chemical with direct evidence in CTD. If no diseases with direct evidence associations are found for the given chemical, an empty list is returned.
ctd_chemical_to_diseases_inferred	Given the name of a chemical, return that chemical's associated diseases with inferred evidence (i.e., from a gene) from the Comparative Toxicogenomics database (CTD). The output from this tool is a list of strings, with each string being of the format: disease_name Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns:

Name	Description
	A list of strings, where each string is a disease associated with the given chemical with inferred evidence in CTD, along with the gene from which the association was inferred. If no diseases with inferred evidence associations are found for the given chemical, an empty list is returned.
ctd_chemical_to_go_biological_process	Given the name of a chemical, return that chemical's associated biological process GO terms from the Comparative Toxicogenomics database (CTD). Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns: A list of strings, where each string is a biological process GO term associated with the given chemical in CTD. If no biological process GO terms are found for the given chemical, an empty list is returned.
ctd_chemical_to_go_cellular_component	Given the name of a chemical, return that chemical's associated cellular component GO terms from the Comparative Toxicogenomics database (CTD). Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns: A list of strings, where each string is a cellular component GO term associated with the given chemical in CTD. If no cellular component GO terms are found for the given chemical, an empty list is returned.
ctd_chemical_to_go_molecular_function	Given the name of a chemical, return that chemical's associated molecular function GO terms from the Comparative Toxicogenomics database (CTD). Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns: A list of strings, where each string is a molecular function GO term associated with the given chemical in CTD. If no molecular function GO terms are found for the given chemical, an empty list is returned.
tox21_assay_predictions	Given the name of a chemical, return that chemical's predicted behavior(s) from Tox21 assays. Each item in the output list contains the assay model name and if the chemical was predicted to be active or inactive. Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters.

Name	Description
	Returns: A list of strings, where each string contains the assay model name and if the chemical was predicted to be active or inactive. The assay model name and activity status are formatted as follows: assay_model_name: active/inactive. If no Tox21 predictions are found for the given chemical, an empty list is returned.
drugbank_genes	Given the name of a chemical, return that chemical's associated gene interactions as documented in DrugBank. This tool returns a list of strings, where each entry is structured like the following: gene_name Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns: A list of strings, where each string is a gene interaction for the given chemical in DrugBank, structured like the following: gene_name
drugbank_atccodes	Given the name of a chemical, return that chemical's therapeutic properties as documented in DrugBank. Use this tool to retrieve a chemical's organ interactions, therapeutic use, and chemical properties. Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns: A list of strings, where each string is a therapeutic property for the given chemical in DrugBank. If no therapeutic properties are found for the given chemical, an empty list is returned.
genra_results	Given the name of a chemical, return corresponding predicted organ interactions, protein interactions, health effects, associated diseases and neoplasticity, developmental toxicity, chronic toxicity, sub-chronic toxicity, subacute toxicity, and reproductive toxicity from the EPA's Generalized Read Across (GenRA) tool. This tool outputs a list of strings. Each string is formatted as follows: category:subcategory - effect Where category may be one of the following: - SAC - Subacute Toxicity - MGR - Multigenerational Reproductive Toxicity - REP - Reproductive Toxicity - CHR - Chronic Toxicity - DEV - Developmental Toxicity - SUB - Sub-chronic Toxicity

Name	Description
	Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns: A list of strings, where each string is a GenRA prediction for the given chemical, formatted as follows: category:subcategory - effect. If no GenRA predictions are found for the given chemical, an empty list is returned.
t3db_targets	Given the name of a chemical, return corresponding targets as documented in the T3DB. This tool can provide information that can help inform how genes and organs interact with a chemical. Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns: A list of strings, where each string is a target associated with the given chemical in T3DB. If no targets are found for the given chemical, an empty list is returned.
toxrefdb_cancer_effects	Given the name of a chemical, return that chemical's toxicological cancer-related effects as reported from studies in ToxRefDB. This tool returns a list of strings, where each item is formatted as follows: effect; toxicity type; species; sex; life stage; target The toxicity type may be one of the following: - ACU - Acute Toxicity - CHR - Chronic Toxicity - DEV - Developmental Toxicity - DNT - Developmental Neurotoxicity - MGR - Multigenerational Reproductive Toxicity - NEU - Neurological - OTH - Other - REP - Reproductive Toxicity - SAC - Subacute Toxicity - SUB - Sub-chronic Toxicity Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns:

Name	Description
	A list of strings, where each string is a cancer-related effect for the given chemical in ToxRefDB, formatted as follows: effect; toxicity type; species; sex; life stage; target. If no cancer-related effects are found for the given chemical, an empty list is returned.
toxrefdb_non_cancer_effects	Given the name of a chemical, return that chemical's toxicological non-cancer-related effects as reported from studies in ToxRefDB. This tool returns a list of strings, where each item is formatted as follows: effect; study type; species; sex; life stage; target The study type may be one of the following: - ACU - Acute Toxicity - CHR - Chronic Toxicity - DEV - Developmental Toxicity - DNT - Developmental Neurotoxicity - MGR - Multigenerational Reproductive Toxicity - NEU - Neurological - OTH - Other - REP - Reproductive Toxicity - SAC - Subacute Toxicity - SUB - Sub-chronic Toxicity Args: chemical_name: A string representing the preferred name of a chemical. This should be a string of at least one character and at most 255 characters. Returns: A list of strings, where each string is a non-cancer-related effect for the given chemical in ToxRefDB, formatted as follows: effect; study type; species; sex; life stage; target. If no non-cancer-related effects are found for the given chemical, an empty list is returned.
structural_similarity	Given a chemical's SMILES representation and a Tanimoto similarity threshold, return structurally similar chemicals with a Tanimoto similarity at or above the specified threshold. Similarity is calculated using Morgan fingerprints. Chemicals returned by this tool may be structurally identical to the input chemical (i.e., synonyms of the input chemical). Args: smiles: A SMILES string representing a chemical's structure. This should be a string of at least one character and at most 255 characters.

Name	Description
	threshold: A float representing the Tanimoto similarity threshold. Unless specified, the threshold is set to a default value of 0.7. Only chemicals with a similarity at or above this threshold will be returned. Returns: A list of strings representing structurally similar chemicals to the original input smiles, where each string is structured as follows: chemical_name
structural_similarity_no_identical	Given a chemical's SMILES representation and a Tanimoto similarity threshold, return structurally similar chemicals with a Tanimoto similarity at or above the specified threshold. Similarity is calculated using Morgan fingerprints. Chemicals returned by this tool cannot be structurally identical to the input chemical (i.e., no synonyms of the input chemical). Args: smiles: A SMILES string representing a chemical's structure. This should be a string of at least one character and at most 255 characters. threshold: A float representing the Tanimoto similarity threshold. Unless specified, the threshold is set to a default value of 0.7. Only chemicals with a similarity at or above this threshold will be returned. Returns: A list of strings representing structurally similar chemicals to the original input smiles, where each string is structured as follows: chemical_name

2.2 Query sets

2.2.1 Basic

This query set consists of 27 questions (or prompts) regarding the overall known or predicted functions and toxicological effects of a chemical on human and rat biology.

2.2.2 Toxicity types

This query set contains the following question for 15 toxicity types (listed below) and toxicokinetic parameters of a chemical on human and rat.

Question

```
List the {effect_type} of {chem_casrn} on human
```

Effect Types

Sub acute, Chronic, Developmental, Immune/lymphatic system, Reproductive system, Nervous system/Neurological, Digestive system, Urinary system, Integumentary system, Musculoskeletal system, Cardiovascular system, Endocrine system, Respiratory system, Hematological system

2.2.3 ABT Q/A

This query set contains the questions from American Board of Toxicology (ABT) certification from the year 2000-2007. These questions have multiple choices of answer. The questions, answer choices and correct answers were extracted from exam sheets of respective years. After filtering ambiguous and unknown answers, the total number of questions (or prompts) is 474. Questions from American Board of Toxicology (ABT) certification exams (2000-20007) were extracted. For a few exams (2005-2007), the answers were provided in tabulated format within a non-pdf file (e.g. MS Excel and MS Word). For the rest, both the questions and answers were provided as scanned pdf format. These pdf files were processed using Amazon Textract which has OCR capability and "Azure GPT-4o" model.

Processing steps of the pdf files

The questions and the answers were processed using the following steps:

Step 1:

- The pdf files were uploaded to Amazon S3, as Amazon Textract does not allow "multi-page" processing for on-premise files
- The relevant pages with questions and answers were filtered and preprocessed using regular expressions
- GPT-4o were used to list the questions and answers (both in string and tabular format) using separate prompts
- Each pdf file was processed separately with the following command

  ```
  python abt-question-ocr/src/extraction_questions.py
  ```

Step 2:

- The pdf outputs were combined with the non-pdf data using the python notebook below

  `abt-question-ocr/notebooks/combine_pdfs.ipynb`

Output format

The output file contains the following columns:

- *Question Index*: Consists of exam year, exam section (or part) and question number. For example, “2000A Q1” denotes the question 1 of the part A of the examination year 2000.
- *Questions*: The question extracted as it was in the pdf
- *Options*: The options extracted as they were in the pdf
- *Question Source*: The source pdf file name that contains the question
- *Answers*: The answer choice(s) for the question
- *Answer Source*: The source pdf file name that contains the answer

Agent focused

This query set contains 10 questions that are not likely to be answered by an LLM without using the RAG step or tools that leverage the internal databases of ToxPipe.

Question	Answer
What human drugs that are structural analogs to CCC(C)(C)C(=O)O[C@@H]1CC@HC?	Simvastatin or Lovastatin or Pravastatin
Identify the chemical is ToxRefDB with the highest structural similarity to [H][C@]12C@HOC(=O)C(C)(C)CC.	Spirodiclofen
What is the Cramer Class of [H][C@]12C@HOC(=O)C(C)(C)CC?	Class III
What enzyme is inhibited by [H][C@]12C@HOC(=O)C(C)(C)CC?	HMGCoA Reductase
What are the most likely side effects of C1CC1C(=O)NC2=CC=CC(=C2)NC3=NC=NC(=C3)NC4=CC=CC(=C4)C(F)(F)F?	Diarrhea, Rash
What is the most likely in vivo molecular/pharmacological target of C1CC1C(=O)NC2=CC=CC(=C2)NC3=NC=NC(=C3)NC4=CC=CC(=C4)C(F)(F)F?	EGFR
Which of these 3 chemicals would be likely lower blood triglycerides: C1=CC(=CC=C1C(=O)/C=C/C2=CC(=C(C=C2)Cl)Cl)OCC(=O)O or CNC1=CC=C(C=C1)O or C[C@]12CC[C@H]3C@HCCC4=CC(=O)CC[C@]34CO?	C1=CC(=CC=C1C(=O)/C=C/C2=CC(=C(C=C
Which of these 3 chemicals has a mode of action that is genotoxic: CCNC1=C(C(=O)C2=C(C1=O)C@HCOC(=O)N)C or C1=CC(=CC=C1C(=O)/C=C/C2=CC(=C(C=C2)Cl)Cl)OCC(=O)O or CNC1=CC=C(C=C1)O?	CCNC1=C(C(=O)C2=C(C1=O)C@HCOC(=O)N

Question	Answer
Which of these 4 chemicals are likely to cross the blood brain barrier and be an effective neuroactive agent: CC1=CC(=C(C=C1)C(C)C)O or CCNC1=C(C(=O)C2=C(C1=O)C@HCOC(=O)N)C or C1=CC(=CC=C1C(=O)/C=C/C2=CC(=C(C=C2)Cl)Cl)OCC(=O)O or CNC1=CC=C(C=C1)O?	CC1=CC(=C(C=C1)C(C)C)O
What chemical is most likely to be exported from cells through MDR1/PGP (ie it is a high affinity subtrate for MDR1): CCC@H[C@@H]1C@HC or C1CCC(CC1)(CC(=O)[O-])C[NH3+] or CN(C)CCO?	CCC@H[C@@H]1C@HC

2.3 Chemicals

The following 20 chemicals were tested with basic and toxicity types prompts.

Chemical	CASRN
Aspirin	50-78-2
Paracetamol	103-90-2
Metformin	657-24-9
Atorvastatin	134523-00-5
Prednisone	53-03-2
Dioxin	1746-01-6
Benzene	71-43-2
Arsenite	15502-74-6
Methylmercury	16056-34-1
Vinylchloride	75-01-4
6PPQ (N-(1,3-dimethylbutyl)-N'-phenyl-p-phenylenediamine)	2754428-18-5
Phenanthrene	85-01-8
TPHP (triphenylphosphate)	115-86-6
Lanthanum	7439-91-0
Galaxolide	1222-05-5
1,6-Dimethyl-3,4-dihydroisoquinoline	91753-09-2
2-Methyl-4-(4-methylphenyl)-2,3-dihydro-1,5-benzothiazepine	74148-62-2
N-Cyclopropylmethyl-10,11-dihydro-5H-dibenzo-(a,d)-cyclohepten-5-imine	59864-46-9
4-Methyl-1,2-dihydrobenzo[f]isoquinoline	29248-42-8
6-Methyl-2,5-diphenyl-6H-1,3,4-thiadiazine	62625-70-1

2.4 Number of queries

The number of queries in an evaluation set is `# base models x # prompts x # chemicals x # species`. For ABT Q/A, this `# base models x # prompts`, as the prompts are not applicable for all chemicals or species.

	Human		Rat		Mixed
	Basic	Toxicity types	Basic	Toxicity types	ABT QA
# prompts	27	15	27	15	474
Base model	2700	1500	2700	1500	2370
RAG	2160	1200	2160	1200	1896
Agentic	2160	1200	2160	1200	1896

3 Evaluation criteria

Not all queries have expected keywords or phrases (or assertions). A response is evaluated by a separate LLM based on semantic similarity with each of the assertions. A response is labeled as "Passed", only if it has semantic similarity with all the assertions.

4 Results

Percentage of successful responses to queries in each evaluation sets by each ToxPipe level. For each evaluation the number of queries with assertion are mentioned in parenthesis.

Model	ABT Q/A (474)	Agent focused Q/A (10)	Tox type prompts (Human) (57)	Tox type prompts (Rat) (12)	Eval Group
Claude 3.7 Sonnet	0.84	0.8	0.82	0.5	Base
Claude 4.5 Haiku	0.8	0.6	0.77	0.39	Base
Claude 4.5 Sonnet	0.85	0.9	0.81	0.59	Base
GPT-4o	0.8	0.6	0.75	0.46	Base
GPT-5 (high reasoning)	0.87	0.7	0.81	0.46	Base
GPT-5 (low reasoning)	0.85	0.6	0.82	0.54	Base
GPT-5 Nano	0.79	0.7	0.65	0.42	Base
Gemini 2.5 Flash	0.85	0.9	0.75	0.56	Base
Gemini 2.5 Pro	0.84	0.9	0.81	0.52	Base

Model	ABT Q/A (474)	Agent focused Q/A (10)	Tox type prompts (Human) (57)	Tox type prompts (Rat) (12)	Eval Group
Llama 4 Scout 17B (Instruct)	0.78	0.2	0.75	0.43	Base
Mistral Large 2	0.7	0.4	0.71	0.44	Base
o3	0.89	0.55	0.82	0.49	Base
Claude 3.7 Sonnet	0.87	0.9	0.78	0.43	MCP
Claude 4.5 Haiku	0.83	0.8	0.61	0.28	MCP
Claude 4.5 Sonnet	0.86	0.9	0.67	0.33	MCP
GPT-4o	0.78	0.6	0.09	0.31	MCP
GPT-5 (high reasoning)	0.87	0.9	0.77	0.17	MCP
GPT-5 (low reasoning)	0.88	0.8	0.79	0.27	MCP
GPT-5 Nano	0.79	0.6	0.62	0.39	MCP
Gemini 2.5 Flash	0.8	0.7	0.19	0.24	MCP
Gemini 2.5 Pro	0.81	0.8	0.62	0.28	MCP
Llama 4 Scout 17B (Instruct)	0.74	0.6	0.02	0.12	MCP
Mistral Large 2	0.64	0.7	0.19	0.1	MCP
o3	0.88	0.8	0.72	0.38	MCP
Toxpipe (RAG) [Claude 3.7 Sonnet]	0.86	0.7	0.77	0.5	RAG
Toxpipe (RAG) [Claude 4.5 Haiku]	0.82	0.6	0.72	0.39	RAG
Toxpipe (RAG) [Claude 4.5 Sonnet]	0.86	0.75	0.77	0.54	RAG
Toxpipe (RAG) [GPT-4o]	0.84	0.4	0.74	0.5	RAG
Toxpipe (RAG) [GPT-5 High Reasoning]	0.88	0.7	0.77	0.45	RAG
Toxpipe (RAG) [GPT-5 Low Reasoning]	0.89	0.6	0.76	0.31	RAG
Toxpipe (RAG) [GPT-5 Nano]	0.77	0.45	0.55	0.31	RAG
Toxpipe (RAG) [Gemini 2.5 Flash]	0.76	0.7	0.67	0.44	RAG
Toxpipe (RAG) [Gemini 2.5 Pro]	0.79	0.55	0.73	0.45	RAG

Model	ABT Q/A (474)	Agent focused Q/A (10)	Tox type prompts (Human) (57)	Tox type prompts (Rat) (12)	Eval Group
Toxpipe (RAG) [Llama 4 Scout 17B (Instruct)]	0.75	0.55	0.69	0.42	RAG
Toxpipe (RAG) [Mistral Large 2]	0.71	0.25	0.38	0.17	RAG
Toxpipe (RAG) [o3]	0.86	0.6	0.76	0.5	RAG